

Synthetic Dataset Design

Graph Data Science — Student Social Network · GDS-26

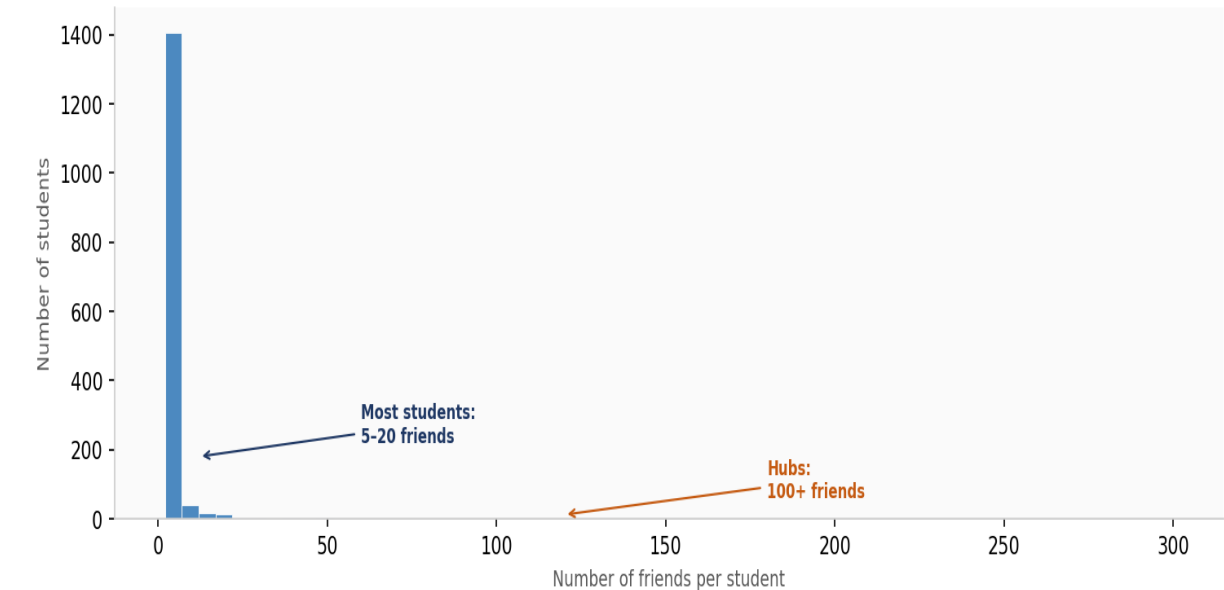
1. Dataset Summary

Element	Decision
Students	1,500
Clubs	70
Events	300
Friendship model	Power-law distribution
Min friends per student	2
Max friends per student	300
Typical range	5 – 20 friends
Hub students	Small % with 100+ friends

2. Friendship Distribution

Friendships follow a **power-law (Zipf) distribution**, meaning a small number of students act as social hubs with many connections, while the majority maintain a modest number of friends. This is consistent with real-world social network topology.

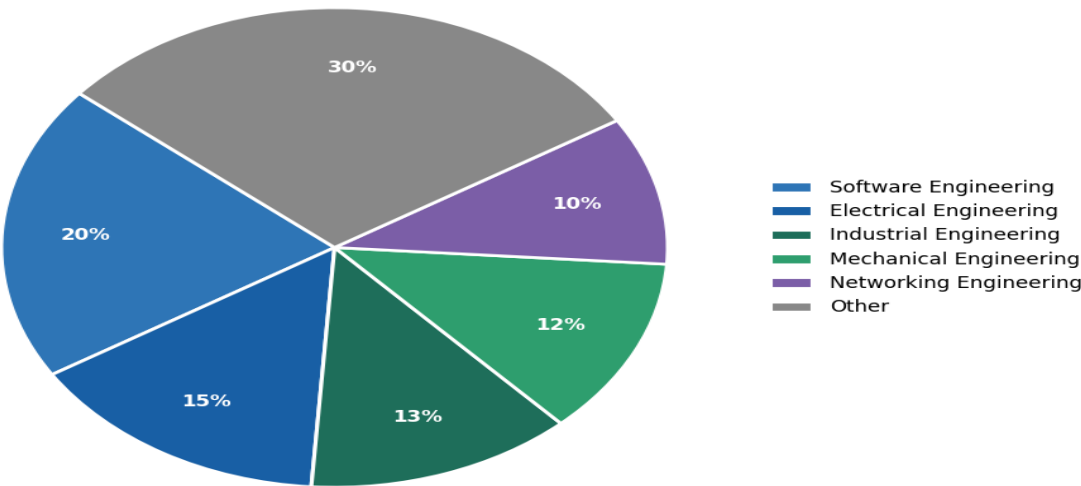
Friendship Degree Distribution (Power-Law)



3. Fields of Study Distribution

Field of Study	Distribution	Students (~)
Software Engineering	20%	300
Electrical Engineering	15%	225
Industrial Engineering	13%	195
Mechanical Engineering	12%	180
Networking Engineering	10%	150
Other	30%	450
Total	100%	1,500

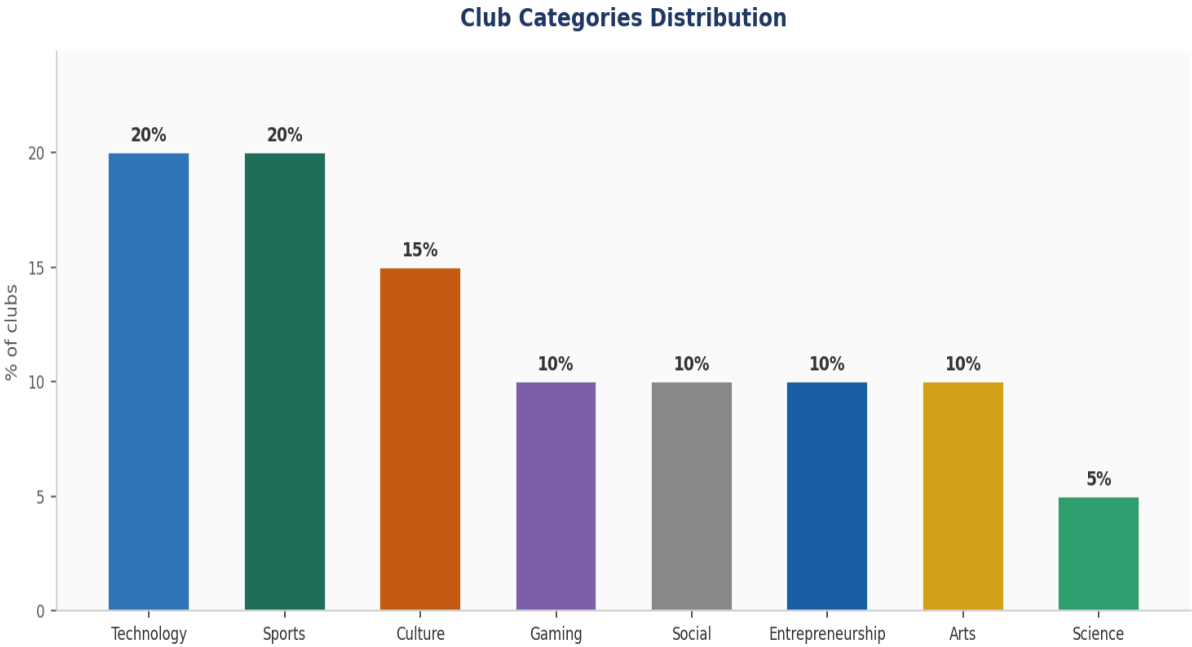
Students by Field of Study



4. Club Categories

Category	Distribution	Clubs (~)
Technology	20%	14
Sports	20%	14
Culture	15%	10
Gaming	10%	7
Social	10%	7
Entrepreneurship	10%	7

Arts	10%	7
Science	5%	4
Total	100%	70



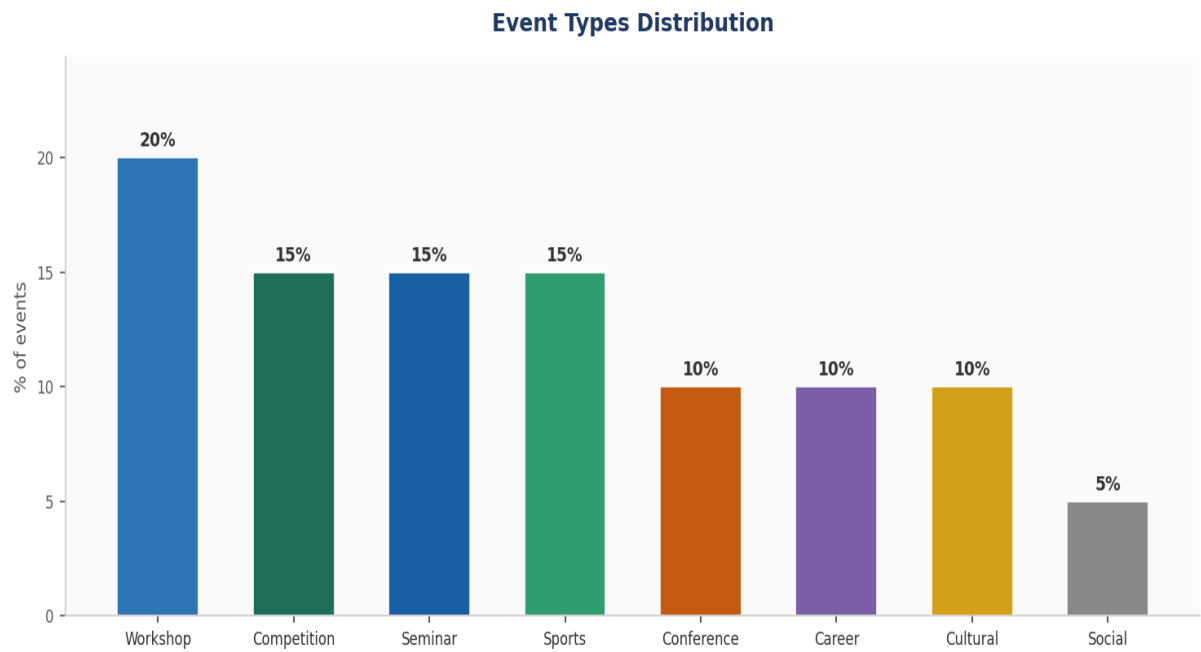
5. Club Membership

A student can join **multiple clubs** with no fixed maximum. The number of clubs per student is generated randomly, weighted so that most students belong to 1–3 clubs, with a small percentage joining more.

6. Event Types

Event Type	Distribution	Events (~)
Workshop	20%	60
Competition	15%	45
Seminar	15%	45
Sports	15%	45
Conference	10%	30
Career	10%	30
Cultural	10%	30
Social	5%	15

Total	100%	300
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7. Event Participation

A student can attend **multiple events** with no fixed maximum. The number of events attended per student is generated randomly. Students who are members of a club have a **higher probability** of attending events organized by that club.